MATH2241 – Introduction to mathematical analysis (1st semester 2526)

Tutorial 3 Solution

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1. Give an example of each of the following, or prove that such a situation is impossible:

   (a) sequences (a_n) and (b_n), where (a_n) converges, (b_n) unbounded and $(a_n b_n)$ converges.

   (b) sequences (a_n) and (b_n), where (a_n) unbounded, (b_n) converges and $(a_n - b_n)$ bounded.

   (c) sequences (a_n) and (b_n), where (a_n) converges, (b_n) diverges and $(a_n + b_n)$ converges.

   (d) sequences (a_n) and (b_n), which both diverge, but $(a_n + b_n)$, $(a_n b_n)$ and (a_n/b_n) all converge.

   Solution.

   (a) $a_n = 0$, $b_n = n$, $a_n b_n = a_n$ converges.

   (b) Let $c_n = a_n - b_n$ be bounded by H, b_n converges, so bounded by some K. Then $|a_n| = |b_n + c_n| \leq |b_n| + |c_n| < H + K$ is bounded.

   (c) Let $c_n = a_n + b_n$ converges, a_n converges, then $b_n = c_n - a_n$ also converges.

   (d) Let $a_n = (-1)^n$, $b_n = -a_n$, then $a_n + b_n = 0$, $a_n b_n = a_n/b_n = -1$ all converges.

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2. If $x_n = \sqrt{n}$, show that (x_n) satisfies $\lim_{n\to\infty} |x_{n+1} - x_n| = 0$, but it is not a Cauchy sequence.

   Solution.

$$\begin{aligned}\lim_{n\to\infty} |x_{n+1} - x_n| &= \lim_{n\to\infty} |\sqrt{n+1} - \sqrt{n}| \cdot \frac{|\sqrt{n+1} + \sqrt{n}|}{|\sqrt{n+1} + \sqrt{n}|} \\ &= \lim_{n\to\infty} \frac{1}{|\sqrt{n+1} + \sqrt{n}|} = 0\end{aligned}$$

   take $\varepsilon = 1 > 0$. For any $N \in \mathbb{N}$, take $h = (N+1)^2 > N$ and $k = N^2 \geq N$, then

$$|x_h - x_k| = |N + 1 - N| = 1 > \varepsilon$$

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3. If $0 < r < 1$ and $|x_{n+1} - x_n| < r^n$ for all $n \in \mathbb{N}$, show that (x_n) is a Cauchy sequence.

Solution.

By triangle inequality, for $n \geq m \geq N$,

$$\begin{aligned} |x_h - x_k| &\leq |x_h - x_{h-1}| + |x_{h-1} - x_{h-2}| + \cdots + |x_{k+1} - x_k| \\ &\leq r^{h-1} + r^{h-2} + \cdots + r^k \\ &< r^k \cdot (1 + r + r^2 + \cdots) \\ &< r^N \cdot \frac{1}{1-r} \end{aligned}$$

Let $\varepsilon > 0$. We want to find $N \in \mathbb{N}$ by Archimedean Property such that

$$r^N \cdot \frac{1}{1-r} < \varepsilon \iff N > \frac{\ln \varepsilon + \ln(1-r)}{\ln r}.$$

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4. If $x_1 < x_2$ are arbitrary real numbers and

$$x_n = (x_{n-2} + x_{n-1})/2 \qquad n > 2$$

show that (x_n) is convergent. What is its limit?

Solution.

Try $x_1 = 0, x_2 = 16$, we see that $x_1 = 0, x_3 = 8, x_5 = 10$ are increasing, $x_2 = 16, x_4 = 12, x_6 = 11$ are decreasing. $x_n \in [x_{n-2}, x_{n-1}]$(or $[x_{n-1}, x_{n-2}]$) and the length of this interval is decreased by 50% each time, so by induction,

$$|x_{n+1} - x_n| = \frac{x_2 - x_1}{2^{n-1}} \quad n \in \mathbb{N}$$

Hence, for $h \geq k \geq N$,

$$\begin{aligned} |x_h - x_k| &\leq |x_h - x_{h-1}| + |x_{h-1} - x_{h-2}| + \cdots + |x_{k+1} - x_k| \\ &= \frac{x_2 - x_1}{2^{h-2}} + \frac{x_2 - x_1}{2^{h-3}} + \cdots + \frac{x_2 - x_1}{2^{k-1}} \\ &< \frac{x_2 - x_1}{2^{k-1}}(1 + 1/2 + 1/4 + \cdots) \\ &\leq \frac{x_2 - x_1}{2^{N-1}} \cdot 2 \leq \frac{x_2 - x_1}{2^{N-2}} \qquad (< \varepsilon \text{ for big } N) \end{aligned}$$

The general term of the subsequence x_{2n+1}:

$$x_{2n+1} = x_1 + \frac{x_2 - x_1}{2}\left(1 + \frac{1}{4} + \frac{1}{16} + \cdots + \frac{1}{4^{n-1}}\right)$$

$$\implies \lim x_n = \lim x_{2n+1} = x_1 + \frac{x_2 - x_1}{2} \cdot \frac{4}{3} = \frac{1}{3}x_1 + \frac{2}{3}x_2$$

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— *End* —